AMR ABOUGHAZALA

Senior Algorithm and Data Processing Developer

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Linkedin Xing profile

EXPERIENCE

Senior Senior Algorithm and Data Processing Developer
Scantinel Photonics GmbH

09/2021 - 30/2025 Ulm, Baden-Württemberg, Germany

- Developed a novel algorithm addressing Doppler ambiguity in FMCW LIDAR, with research on recent publications into practical implementations with CARLA-based simulation.
- Implemented an internal perception pipeline with point cloud novel filtering algorithms, Ransac segmentation, kNN, KD-tree for object detection and Kalman tracking.
- Implemented PointNet and PV-RCNN in PyTorch, building a Docker-based development environment.
- Developed pixel-level statistical analysis tools to evaluate different filtering and peak detection algorithms on raw time-base data.
- Maintained real-time LiDAR processing algorithms ensuring reliable performance between ARM processor and host with PTP synchronization.
- Implemented a real time visualizer PyQt with MVC pattern, integrating the SW backend API with foxglove.

Software Developer
Navimatix GmbH

01/2019 - 07/2021 Jena, Thuringen, Germany

- Implemented an image processing pipeline for bacteria counting on fluorescence microscopy images E. coli, P. aeruginosa, M. tuberculosis, S. aureus), integrating ImageJ/FIJI APIs in a JavaFX GUI.
- The workflow applied Median filtering, ISODATA segmentation, Opening filters, and 2D/3D object counting using convex-acute angles to separate overlapping Bacteria.
- Implemented several user interface applications in Java JavaFX , C# (.Net Framework WPF) and Delphi

EDUCATION

M.Sc. in Communications and Signal Processing
Technische Universität Ilmenau

09/2014 - 02/2018 Ilmenau, Thuringen, Germany

- Major Subjects: Mobile Communications Engineering, Adaptive Array and Signal Processing, Digital Signal Processing and Communication Networks (Avg. Grade 2.2/1.0).
- Thesis: 'Positivity Decomposition Algorithms on EEG/MEG Data' Implemented non-negativity constraints on tensor-based (Grade 1.3/1.10).

B.Sc. in Electronics and Communication
Arab Academy for Science and Technology

09/2005 - 02/2010 Alexandria, Egypt

- Major Subjects: Wireless and Mobile Communications, Analog and Digital Signal Processing, Communication Networks, Electro-Magnetic and Antenna theories (Avg. Grade 1.56/1.0).

SKILLS

Signal Processing • Algorithm Development • Machine Learning • Sensor Fusion
Image Processing • 3D Point Clouds • Optimization • Wireless Communication

Python • MATLAB • C++ • Java • C
PyTorch • NumPy • SciPy • scikit-learn • OpenCV
Open3D • Open3D-ML • PyTorch3D • CARLA • Docker • Git

Languages

Arabic: Mother Tongue
English: Fluent
German: B1 – TELC